



2IA Practice Problems I

Integers



Practice Problems Chapter 1

1. **Prove each statement by induction on n .**

(a) $1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$, for all $n \geq 1$.

(b) $\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n-1}{n!} = \frac{n!-1}{n!}$, for all $n \geq 2$.

(c) $\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \dots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n}$, for all $n \geq 2$. (It was in Test 1, 2021)

(d) $(-1)^1 \cdot 1^2 + \dots + (-1)^n \cdot n^2 = \frac{1}{2}(-1)^n \cdot n(n+1)$, for all $n \geq 1$. (It was in Test 1, 2022)

2. Let a_n be the sequence given by $a_1 = 1$, $a_2 = 8$ and $a_n = a_{n-1} + 2a_{n-2}$ for $n \geq 3$.
Using strong induction prove that

$$a_n = 3 \cdot 2^{n-1} + 2(-1)^n,$$

for all $n \in \mathbb{N}$.

3. **Apply the Euclidean Algorithm to compute $\gcd(7854, 2145)$ and express it as a linear combination of 7854 and 2145.**

4. **In each case determine whether the statement is true or false.** If true, provide an argument supporting your answer; if false, provide a counterexample. ((a)-(c) were in Test 1, 2021.)

(a) Suppose that n and m are two integers not both zero, and $d > 1$ is an integer satisfying that $d = xm + yn$ for some integers x and y . Then $d = \gcd(n, m)$.

(b) Suppose that n and m are two integers not both zero such that $1 = xm + yn$ for some integers x and y . Then $1 = \gcd(n, m)$.

(c) If $d = \gcd(n, m)$ and $n = da$ and $m = db$. Then $\gcd(a, b) = 1$.

(d) If $a = a_1c$ and $b = b_1c$, where $c = \gcd(a, b)$, then $\text{lcm}(a, b) = a_1b_1c$.

5. **Use the Prime Factorisation Theorem to find the gcd and the lcm of the following pair of numbers:**

(a) $m = 139$, $n = 278$

(b) $m = 735$, $n = 110$

6. Suppose that m and n are coprime and that $k|m$. Prove that k and n are coprime.

7. **In each case determine whether the statement is true or false.** If true, provide an argument supporting your answer; if false, provide a counterexample. ((e) was in Test 1, 2021, while (f) was in Test 1, 2022.)
- (a) $40 \equiv 13 \pmod{9}$
 - (b) There are no integers a and b such that $a^2 - 3b = 2$.
 - (c) $(51 + 68) \equiv 1 \pmod{5}$.
 - (d) $1111 \equiv 1 \pmod{2}$.
 - (e) Suppose that $p > 2$ is a prime. If $p \equiv 3 \pmod{4}$, then $(p - 1)/2$ is odd.
 - (f) No integer of the form $n^2 + 1$ is a multiple of 7.
8. Find the remainder of 6^{1001} when is divided by 7 (it was in Test 1, 2021).
9. (a) **In $\mathbb{Z}_{20}$, find the inverse of $\overline{11}$ and use it to solve $\overline{11}x = \overline{16}$.**
 (b) **In $\mathbb{Z}_{15}$, find the inverse of $\overline{4}$ and use it to solve the equation $\overline{4}x = \overline{12}$.**
 (It was in Test 1, 2021).
10. Consider the decimal representation $n = d_j d_{j-1} \dots d_2 d_1 d_0$ of a positive integer n . Prove that $3|n$ if and only if $3|(d_0 + d_1 + \dots + d_j)$. (It was in Test 1, 2022.)
- Just in case you need an example of the decimal representation: for $n = 3785$ we have that $d_0 = 5$, $d_1 = 8$, $d_2 = 7$ and $d_3 = 3$.
11. Find all the solutions to the simultaneous congruences: ((b) was in Test 1, 2022)
- (a) $x \equiv 1 \pmod{5}$ and $2x \equiv 1 \pmod{9}$
 - (b) $3x \equiv 2 \pmod{5}$ and $x \equiv 2 \pmod{3}$